

Prediction of Diabetes Patients Using a Multi-Layer Perceptron Neural Network

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Abstract

Early identification of diabetes risk supports timely clinical intervention and improved patient outcomes. This paper presents a Multi-Layer Perceptron (MLP) neural network for binary classification of diabetes status from eight routinely collected clinical features: pregnancies, glucose, blood pressure, skin thickness, insulin, body mass index (BMI), diabetes pedigree function, and age. Exploratory analysis identifies physiologically implausible zero values in several features, which are corrected by mean imputation, while class-wise interquartile-range filtering removes outliers from the two most predictive features, glucose and BMI. The cleaned data are standardised and split into stratified training, validation, and test sets. An MLP with two hidden layers, ReLU activations, dropout regularisation, and a weighted binary cross-entropy loss is trained with the Adam optimiser and early stopping. Three architectural configurations are compared on a held-out validation set using accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC). A configuration with two hidden layers of 128 and 64 units achieves the highest validation AUC (0.870) and is selected as the final model. On the unseen 155-sample test set, this model attains an accuracy of 0.745, a recall of 0.737, and an AUC of 0.848, indicating reliable probability ranking despite a modest reduction in point accuracy relative to the smallest configuration. These results suggest that, for clinical screening tasks where missing a positive case carries a high cost, model selection by AUC rather than accuracy alone yields a more clinically appropriate classifier.

Keywords: Diabetes Prediction; Multi-Layer Perceptron; Neural Networks; Clinical Decision Support; Binary Classification; Class Imbalance; Area Under the Curve

1 Introduction

Diabetes prediction is an important task that supports the identification of individuals at risk of developing serious complications. Early prediction plays a crucial role in enabling timely medical intervention and improving patient outcomes (World Health Organization, 2023). In this study, a real-world clinical dataset is used to classify whether a patient has diabetes (Outcome = 1) or not (Outcome = 0) on the basis of eight clinical features — Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age — that are commonly associated with diabetes risk. The dataset is divided into a training pool of 615 records and a held-out test set of 155 records. The goal is to train a Multi-Layer Perceptron (MLP) to learn patterns from the training data and to predict diabetes status accurately on the unseen test records.

The remainder of this paper is organised as follows. Section 2 reviews related work on machine learning for medical diagnosis and diabetes prediction. Section 3 describes the dataset, the exploratory analysis and preprocessing pipeline, the MLP architecture, and the training and evaluation protocol. Section 4 presents the validation comparison across three MLP configurations, reports the final model’s performance on the test set, and discusses the results. Section 5 concludes the paper and outlines directions for future work.

2 Related Work

Machine learning has been applied widely to medical diagnosis, where the goal is typically to support rather than replace clinical judgement; Sharma et al. (2021) survey this application area and highlight the importance of matching model choice to data characteristics and clinical requirements. Within diabetes prediction specifically, Chakraborty and Chatterjee (2019) compare several classical machine learning techniques, while Reddy et al. (2020) analyse a range of algorithms for diabetes risk estimation, both finding that careful preprocessing and feature selection materially affect predictive performance. More closely related to the present work, Rahman, Bashar, and Moni (2021) use deep learning models on clinical and lifestyle features to predict diabetes, and Sørensen, Dam, and Nielsen (2022) directly compare neural networks against logistic regression for type-2 diabetes prediction, reporting that neural architectures can offer improved discrimination at the cost of interpretability. Zhou, Chen, and Li (2021) review deep neural network architectures for medical classification problems more broadly, noting that architecture and regularisation choices (depth, width, dropout) strongly influence generalisation on the comparatively small, tabular datasets typical of clinical applications — a consideration that directly motivates the architecture search performed in this paper.

The neural network design and training practices used here draw on standard deep learning methodology, including rectified linear activations, dropout regularisation, and adaptive gradient-based optimisation (Goodfellow, Bengio and Courville, 2016). Data preprocessing choices, including outlier handling via the interquartile range and z-score standardisation, follow established data mining and statistical learning practice (Han, Kamber and Pei, 2012; Friedman, Hastie and Tibshirani, 2009; Kotu and Deshpande, 2019).

3 Materials and Methods

3.1 Dataset Description

The dataset comprises clinical records described by eight numeric features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, and Age, together with a binary Outcome label indicating diabetes status. The data are provided as a training pool of 615 records and a held-out test set of 155 records; the test set is not used at any stage of model development or selection.

3.2 Exploratory Data Analysis and Preprocessing

Prior to model development, exploratory data analysis was performed using histograms and boxplots to examine the distribution of each feature and to identify abnormal values (Figure 1). This analysis showed that Glucose, Blood Pressure, Skin Thickness, and BMI contained values of zero, which are not physiologically possible for these measurements. Glucose and BMI contained 4 and 11 zero values respectively, while Blood Pressure and Skin Thickness each contained 211 zero values. Deleting 211 rows would have drastically reduced the size of the dataset and increased the risk of overfitting, so the zero values in Blood Pressure and Skin Thickness were instead replaced with the mean of the corresponding non-zero values.

A correlation heatmap (Figure 2) was also examined, showing that Glucose and BMI were the two features most strongly associated with the diabetes outcome. These two features were therefore subjected to class-wise Interquartile Range (IQR) outlier removal, preserving the overall size of the dataset while limiting the influence of extreme values on the most predictive features. Figures 3 and 4 show the resulting feature distributions and outcome-stratified boxplots after cleaning.

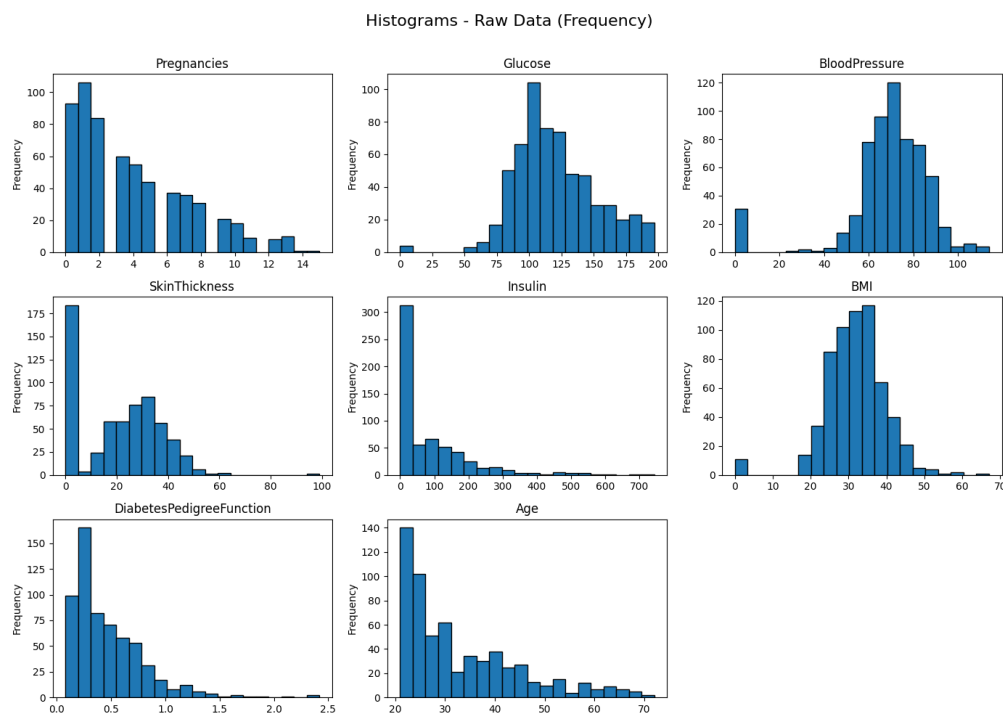


Figure 1: Histograms of the eight clinical features in the raw training data.

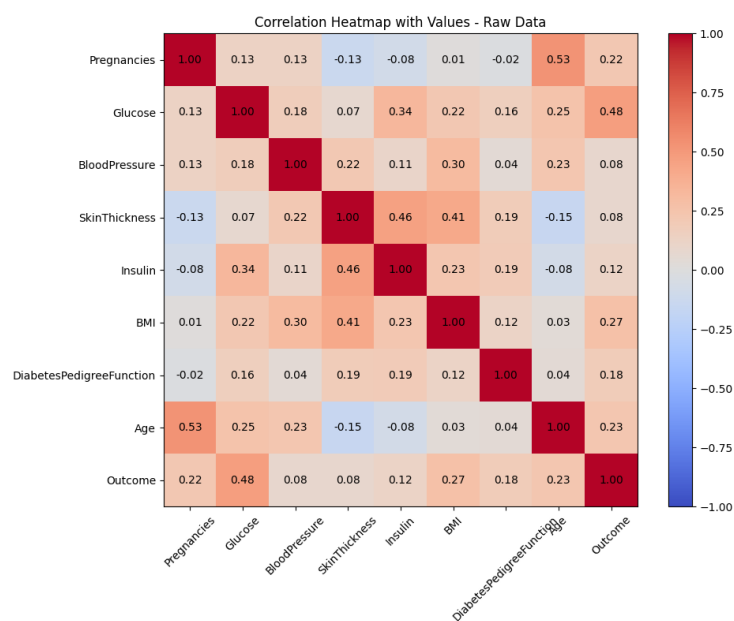


Figure 2: Correlation heatmap of the eight clinical features and the diabetes outcome (raw data).

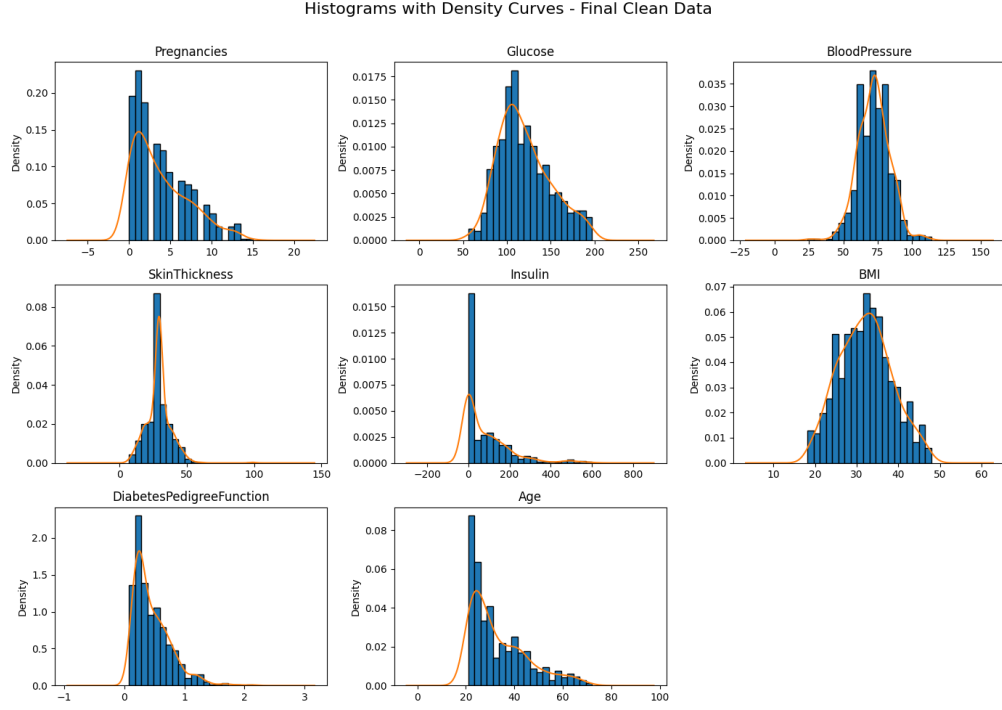


Figure 3: Histograms with fitted density curves for the eight clinical features after cleaning.

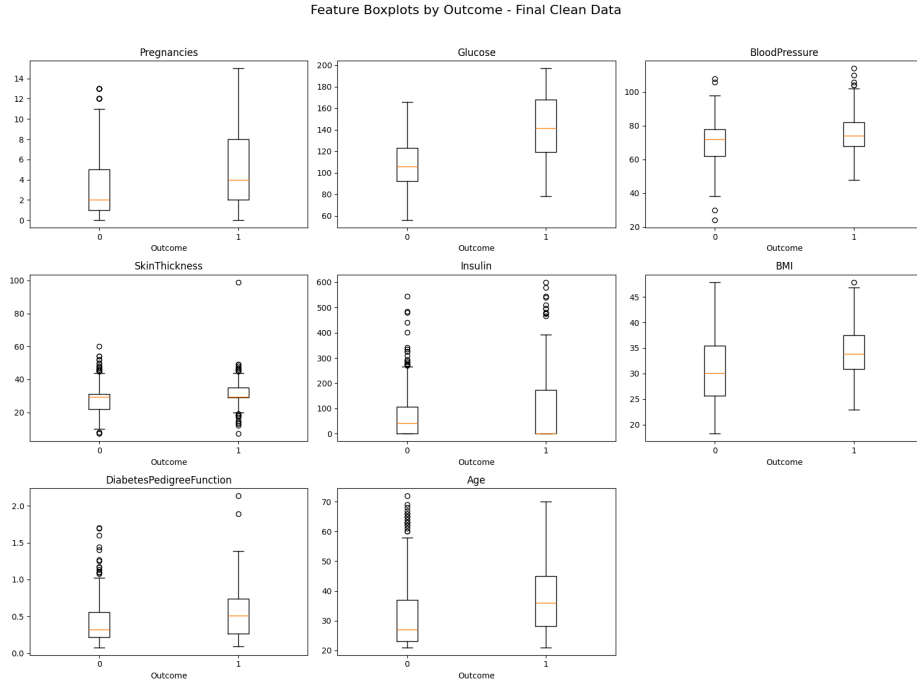


Figure 4: Boxplots of each feature stratified by diabetes outcome, final cleaned data.

Finally, all eight features were standardised using z-score normalisation, and the cleaned dataset was divided using stratified sampling into an 80%/20% split to preserve class balance. After cleaning, 579 of the original 615 training-pool records remained; stratified splitting of this cleaned pool yielded 463 training samples and 116 validation samples, with the original 155-record test set held out unchanged throughout preprocessing and model development.

3.3 Network Architecture

A Multi-Layer Perceptron (MLP) was implemented in PyTorch for binary diabetes classification. The network consists of an input layer of eight neurons (one per feature), two fully connected hidden layers, and a single-neuron output layer. Each hidden layer uses a Rectified Linear Unit (ReLU) activation, chosen for its faster convergence and its avoidance of the vanishing-gradient problem, and is followed by dropout regularisation to reduce overfitting by randomly deactivating neurons during training. The output layer produces a single raw score (logit), which is converted to a probability by a sigmoid function during evaluation.

3.4 Training Configuration

The model was trained with the Adam optimiser using a learning rate of 0.001 as the baseline setting, with the optimiser adjusting weights adaptively during training. The loss function was Binary Cross-Entropy with Logits (`BCEWithLogitsLoss`), which combines a sigmoid activation with the cross-entropy loss in a single numerically stable operation. Because the two outcome classes were moderately imbalanced, a class weight was applied to the loss so that misclassifying a diabetic patient was penalised more heavily, improving the model’s sensitivity to positive cases. Training used mini-batches of size 32, and early stopping with a patience of 10 epochs was applied to prevent overfitting, halting training once validation performance ceased to improve.

3.5 Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (AUC). Accuracy reflects overall correctness; precision measures the proportion of predicted-diabetic cases that are true positives; recall measures the proportion of actual diabetic cases correctly identified; and the F1-score balances precision and recall. AUC summarises the model’s ability to rank patients by risk across all possible classification thresholds and is particularly important in medical applications, where failing to identify a diabetic patient can have serious consequences.

3.6 Experimental Design

Three MLP configurations were evaluated to assess the effect of hidden-layer size, dropout rate, and learning rate on predictive performance: a small network, a baseline network, and a large network, described further in Section 4. All three configurations were trained under the same protocol described above — mini-batch size of 32, Adam optimisation, class-weighted loss, and early stopping — to ensure that observed differences in performance reflect architectural and hyperparameter choices rather than training procedure.

4 Results and Discussion

4.1 Validation Performance Comparison

Three MLP configurations were compared by varying the hidden-layer sizes, learning rate, and dropout rate. Table 1 summarises the results on the stratified validation set, and Figure 5 shows the corresponding training and validation loss curves for each configuration.

Table 1: Validation results for the three MLP configurations.

Model	Hidden	LR	Dropout	Acc	Prec	Recall	F1	AUC
Exp1_Small	32 → 16	0.01	0.2	0.7931	0.6429	0.9000	0.7500	0.8655
Exp2_Baseline	64 → 32	0.001	0.2	0.7845	0.6415	0.8500	0.7312	0.8632
Exp3_Large	128 → 64	0.0005	0.3	0.7759	0.6346	0.8250	0.7174	0.8704

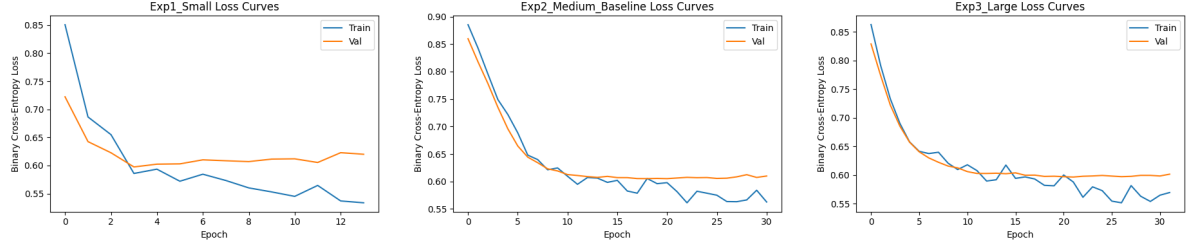


Figure 5: Training and validation loss curves for Exp1_Small (left), Exp2_Baseline (centre), and Exp3_Large (right).

Experiment 3 achieved the best AUC (0.8704) of the three configurations, while its accuracy, precision, recall, and F1-score remained broadly comparable to the other two models. AUC is the most clinically significant metric in this setting, as it measures the model’s ability to discriminate between diabetic and non-diabetic patients across all probability thresholds, independent of a single fixed decision boundary. This indicates that Experiment 3 provides the most reliable probability estimates and the strongest discriminative power, both prerequisites for clinical decision support. Although Experiment 1 achieved a slightly higher accuracy and recall, its lower AUC implies weaker risk ranking and less flexibility in threshold selection — a limitation illustrated by the fact that a high recall is of limited value if the underlying probability estimates are poorly calibrated. Figure 6 shows the confusion matrix and ROC curve for Experiment 3 on the validation set; the ROC curve’s consistent separation from the diagonal chance line supports Experiment 3’s selection as the most suitable model for diabetes screening, where missed diagnoses are particularly undesirable.

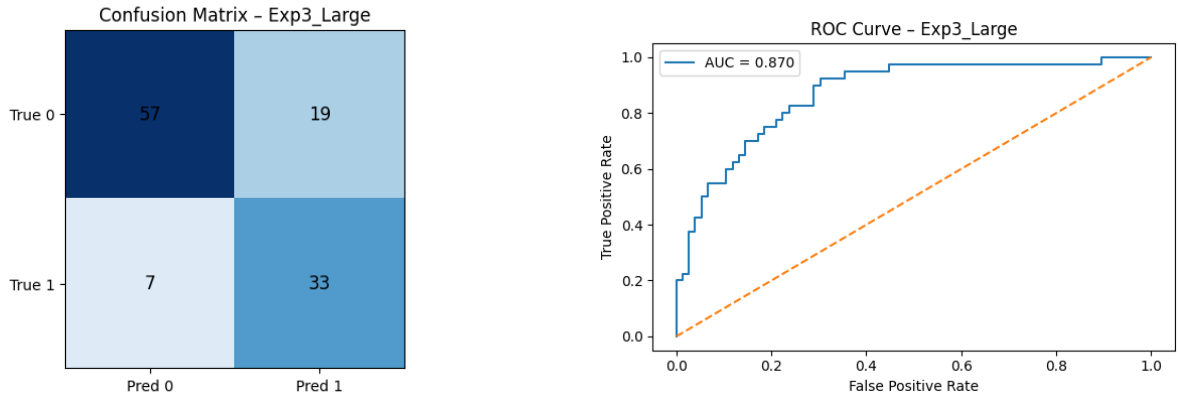


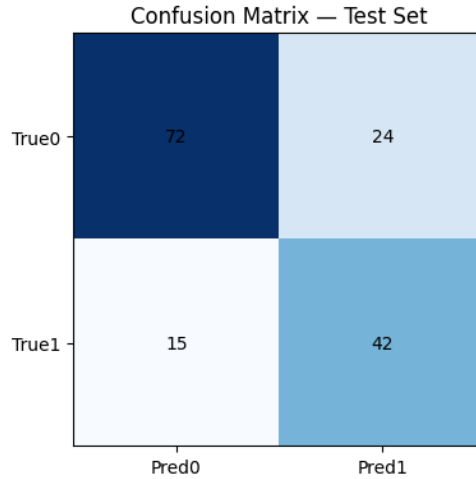
Figure 6: Confusion matrix (left) and ROC curve (right) for Exp3_Large on the validation set.

4.2 Final Model Selection and Test Set Evaluation

On the basis of its superior validation AUC, Experiment 3 (the large configuration) was selected as the final model due to its stronger clinical prioritisation properties. This model was then evaluated once on the previously unseen 155-sample test set. Table 2 reports the resulting metrics, and Figure 7 shows the corresponding confusion matrix.

Table 2: Test-set performance of the selected final model (Exp3_Large).

Metric	Test Score	Interpretation
Accuracy	0.7451	Good overall classification
Precision	0.6364	Acceptable false-positive balance
Recall	0.7368	Strong diabetic detection
F1-score	0.6829	Balanced performance
AUC	0.8480	High reliability in probability ranking

**Figure 7:** Confusion matrix for the final model (Exp3_Large) on the held-out test set.

4.3 Discussion

The evaluation results indicate that AUC is the most valuable measure of clinical usefulness in this task, since it captures the consistency and reliability of separation between high-risk and low-risk patients across all decision thresholds. By this measure, Experiment 3 showed the highest predictive confidence and the best generalisation from validation to test data. Although its accuracy and F1-score were somewhat lower than those of the smallest configuration, its performance was stable across both the validation and test sets, with a validation AUC of 0.870 and a test AUC of 0.848. The test-set recall and AUC (0.7368 and 0.8480 respectively) show that the final model correctly identifies the majority of diabetic patients while maintaining reliable probability ranking, supporting its selection as the most clinically appropriate of the three architectures evaluated.

5 Conclusion

This paper presented an MLP-based approach to diabetes prediction from eight routine clinical features, combining principled handling of physiologically invalid zero values, class-wise outlier removal on the most predictive features, and a class-weighted training objective to address moderate class imbalance. Three MLP configurations, varying in hidden-layer size, learning rate, and dropout, were compared on a stratified validation set using accuracy, precision, recall, F1-score, and AUC. The largest configuration (128 $\rightarrow$ 64 hidden units) achieved the highest validation AUC (0.870) and was selected as the final model, going on to achieve an AUC of 0.848 and a recall of 0.737 on the previously unseen test set, despite a modest reduction in point accuracy relative to the smallest configuration. These findings support the use of AUC, rather than accuracy alone, as the primary model-selection criterion in clinical screening applications, where the cost of a missed diagnosis is high. A limitation of the present study is that

hyperparameters were varied jointly across configurations rather than through a systematic grid search, so the individual contribution of hidden-layer size, learning rate, and dropout cannot be fully disentangled; future work could address this through a controlled ablation study, external validation on an independent clinical cohort, and calibration analysis of the predicted probabilities to further support clinical decision-making.

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